

TOUCHE: AN AFFORDABLE LIFE-CHANGER

For The Visually-Challenged



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The world is becoming digital. We wake up every morning on seeing the time on our clock/watch. We turn on the television or e-readers and catch up on news. We leave for work, go to the station and the display tells us how long before the train arrives. At office, we spend most of the day working on our laptop. Let us not forget that the whole day we have a companion with us, the smartphone. Imagine if, just for a day, you had to stay digital-less.

“No WhatsApp! No browsing! No e-books!” Just reading this, I am sure, must have given you the creeps. We do not even give a second thought to these tasks that have become a way of life today. But there are people out there who need someone to read the digital data out loud; they need special books printed for them to read content that is always outdated and a scribe for every examination.

The question that prompted it all

Is there any way to make their life easier, to make them technology-enabled? Those in the know might say that there are

What is Touche

Touche is a stand-alone e-reader and note-taker combined, and a mini-computer for the visually-challenged. It converts digital information (ASCII) into tactile (feel-able) Braille. Apart from its own keyboard for data entry, contents of your smartphone, computer or memory stick can be fed to the device. It mechanically produces Braille dots that change according to the information being received. Touche can also output information to a Braille embosser if a hard copy is needed, or send information back to a phone or a computer.

devices that do the job. Yes, there are and these are made by foreign companies using German or Japanese technology, or by Indian companies that borrow their technology. No wonder then that the already overburdened visually-challenged find the US\$ 2000 per line Braille converters unaffordable. Can we find an alternative to this? This is the question Paul Gerard D'Souza, a freelancer who invents new technologies and devices, set out to answer.

We know the problem, what do we do

We need a device that can take in digital text and convert it to Braille. The first part is easy—use a computer or smartphone to type out the content. Transfer this data to the device via Bluetooth, USB or a pen-drive and throw in some memory to store it. D'souza used customised software to convert this digital text into signals, and a processor to control this.

Let us now take a look at the other side. The display demands a tactile surface on which small projections are made to appear and recede, giving the user the Braille reading interface.

Braille display matrix has fixed dimensions, and different dot combinations

Touche e-reader

